Thangatamil A

Software Development Engineer

LinkedIn — GitHub — Portfolio — thangatamil1177@gmail.com — +919025565212

PROFILE

Software Development Engineer with 2.5+ years of hands-on experience in orchestrating, designing and developing software solutions. Handling client communications in technical discussions. Working effectively in Agile development environments. Reviewing codebases and Reverse engineering legacy architectures. Specialized in building scalable microservices, and real time communication platforms with a focus on performance, reliability and distributed system with high availability.

EXPERTISE

- **Frontend & Web**: Html, Javascript, React.js, Vite.js.
- **Backend** & **Messaging:** Java Spring Boot, Golang, Erlang/otp, Ejabberd.
- **Protocols:** Http, WebSocket, Tcp, Udp, Xmpp, Smtp.
- **S2S Communication:** Rest APIs, Grpc Protobuf, RabbitMQ, Kafka.
- **Databases** & **Caching:** MySQL, SQLite, PostgreSQL, MongoDB, Redis.
- **Architectures** & **Methodologies:** Monolithic, MicroService. Agile, Scrum, Waterfall.
- **Cloud** & **DevOps:** AWS, Oracle, Minio (on-premises), GCP. Docker, Kubernetes, CI/CD, Jenkins.
- **Tools** & **LLM:** Jira, Grafana, Postman, Git, Bitbucket, Github, Psi. OpenCode, Claude, Cursor.

EXPERIENCE

Contus Tech Sep 2024 – June 2026

Designation: *Junior Software Development Engineer*

Product: **Mirrorfly**

- Replaced **single-service API Gateway** dependency with highly available **Nginx/Apache proxy** servers, improving gateway availability to **99.99%** and reducing gateway-related latency by **20%**. Developed a reusable Maven JWT authentication library adopted across **10+** Spring Boot microservices.
- Rewrote the **Contact synchronization** microservice from **Java** to **Go**, overcoming **JVM heap limitations** for large datasets. Reduced database load through Redis caching and introduced batched, parallel synchronization strategies to improve scalability.
- Modernized the microservice architecture by migrating from **Spring MVC + Tomcat** to **Spring WebFlux + Netty** for non-blocking I/O and advocated adoption of **R2DBC** over traditional JDBC/JPA for reactive database access.
- Re-architected the **Ejabberd authentication pipeline** by replacing external **Go/Python authentication** services with JWT-based authentication backed by Redis session validation. Reduced authentication latency from **200ms–3s to 1–2ms**, enabled multi-session login through schema redesign, resolved database deadlocks, and introduced scheduled cleanup of expired JWT tokens.
- Designed a horizontally scalable **Redis caching and persistence** architecture on Kubernetes to offload SQL traffic by caching user presence, blocking status, mute configurations, roster data, and group participants, substantially improving real-time messaging performance.
- Integrated the **Firebase Admin SDK** into the backend to validate **Firebase Authentication** ID tokens during user registration, ensuring secure user verification before account creation.
- Introduced **gRPC** with **Protocol Buffers** for internal service-to-service communication, significantly reducing serialization overhead and improving communication performance.

Product: **Hike Messenger**

- Eliminated the **Java XMPP microservice** by developing a native **RabbitMQ module** in ejabberd with asynchronous worker pools for **publish/subscribe** processing, reducing infrastructure overhead, removing inter-service network calls, and improving message throughput.
- RabbitMQ module: Designed using **Erlang's philosophy** of **Let it crash** and **fault tolerance**. **Worker pool** for rabbitMQ consumer process to handle medium to large scale data processing.
- Orchestrated the **message reactions** feature for one-to-one and group messaging using ejabberd. Implemented asynchronous database operations, offline synchronization using Firebase Cloud Messaging (**FCM**) and Apple Push Notification service (**APNs**), and reliable persistence (**spool/offline storage**) as a fallback mechanism.
- Worked with **Erlang/OTP concurrency** and **fault tolerance** using **lightweight processes**, asynchronous message passing, **mailboxes**, pattern matching, and **supervision trees** for **process monitoring** and automatic recovery.
- Solved Ejabberd's core sql module's **P1_fsm behavior** for transaction **Re-queuing** and **Replaying** on TCP network connection failure states. Ensuring zero transaction failure metrics for data consistency.
- Optimized high-volume MySQL workloads through native query tuning, **Parameterized queries, Composite indexing** and **Transactional consistency**, significantly improving message-processing efficiency while ensuring reliable rollback behavior.
- Worked extensively with real-time communication technologies including Ejabberd, RabbitMQ, Redis, Spring WebFlux, Go, gRPC, Protocol Buffers, Janus, SIP, LiveKit (WebRTC), and Centrifugo.

Innov Source — Tech Mahindra's client Apr 2024 – Sep 2024
Designation: *FTTX Engineer*

- Developed a strong foundation in binary data representation and how digital information is transmitted across networks using **electromagnetic signals** and **fiber-optic communication** in cellular and Ethernet environments.
- Gained hands-on understanding of how **public** and **private IP addresses**, NAT, and port mapping work together to route packets to specific devices and processes.

PRODUCT EXPERIENCE

Products I have dealt with:

- Core: **MirrorFly, Hike Messenger.**
- Forked: **Ejabberd XMPP, LiveKit WebRTC, Centrifugo TCP/WebSocket.**

CERTIFICATIONS

- **Spring Boot** — AmigosCode **view** Aug 2023 – Mar 2024
  Hands-on course focused on building RESTful APIs in microservices.
- **Java: Spring** — QSpiders
  Comprehensive training covering Core Java, OOP, collections, multithreading, exception handling.
- **MERN Stack** — EMC **view**
  Focused on building modern web apps and integrating REST APIs using React, Node.js and mongoDB.

EDUCATION

Bachelor of Engineering — Computer Science and Engineering Jun 2019 – May 2023
Hindusthan Institute of Technology, CGPA: 7.8